PAPER SUMMARY

# AI-Driven Autonomous Infrastructure Optimization for Large-Scale Distributed Systems

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## ABSTRACT

Large distributed systems rely on independent control loops—autoscalers, provisioning systems, and routing managers—each optimizing in isolation. The result is conflicting decisions, wasted resources, and latency SLO violations.

This paper presents GA-AOCP (Graph-Aware Autonomous Optimization Control Plane), a system that models infrastructure as a time-evolving dependency graph and uses constrained reinforcement learning to generate coordinated optimization decisions across scaling, provisioning, and routing simultaneously. The problem is formulated as a constrained Markov Decision Process (CMDP), enabling multi-objective optimization across cost, performance, and carbon impact while enforcing hard service-level boundaries.

Evaluation on microservice benchmarks and cluster traces shows a 23 percentage point CPU utilization gain over rule-based HPA, 19% cost reduction, a drop in p99 SLO violations from 1.8% (rule-based HPA) to 0.4%, and 14% carbon reduction, while eliminating action conflicts entirely. GA-AOCP demonstrates that unified, graph-aware infrastructure optimization is technically feasible and operationally practical.


## HIGH-LEVEL ARCHITECTURE

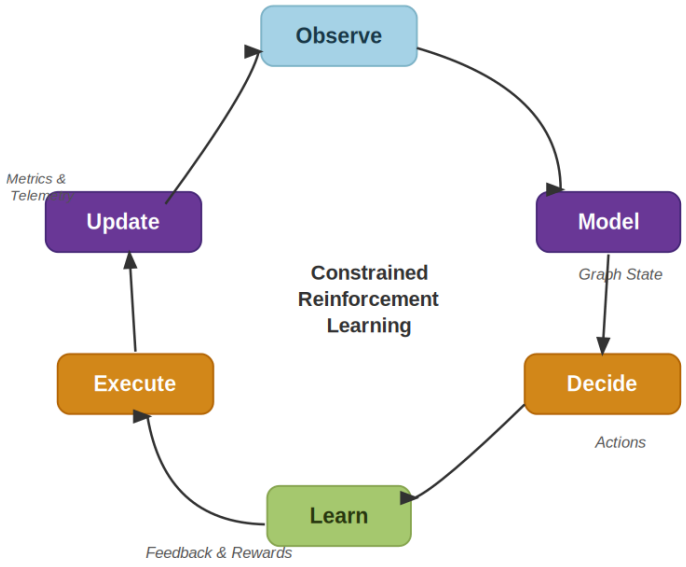



**Fig. 1.** GA-AOCP control loop: continuous observation, graph-based modeling, constrained RL decision-making, execution, and feedback-driven learning.

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